

Harsh Tiwari

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EXPERIENCE

WellnessLiving

Senior Software Engineer

Bengaluru, India

April 2025 - Present

- **Voice AI Agent Platform:** Built a multi-tenant Voice AI backend and designed a generalized function-calling layer to orchestrate LLM-driven actions across diverse provider integrations, enabling extensible, tenant-specific workflows with RAG and tool execution pipelines.
- **Low-Latency LLM Serving for Telephony:** Re-architected Cerebrum, our LLM microservice, for sub-second telephony responses by streaming model output, parallelizing retrieval and tool preparation, trimming prompt construction overhead, and removing synchronous bottlenecks from the inference path.
- **Real-Time Chat Infrastructure:** Designed a horizontally scalable WebSocket platform with stateless services, JWT authentication, and rate limiting, improving connection stability and reducing customer-reported issues.
- **Data Ingestion Pipeline:** Built Airflow-based ingestion pipelines from systems including Intercom and Zoom into Redshift, improving reliability of centralized analytics and downstream reporting.
- **Agentic Analytics Interface:** Built a RAG-backed analytics interface with SQL agents to support self-serve reporting for internal teams.
- **Latency-Critical Voice Pipeline Observability:** Built stage-level latency instrumentation across STT, request assembly, LLM inference, tool execution, and TTS to attribute tail latency in the telephony path and guide sub-second response optimizations in production voice systems.

Kusho

Full Stack Developer

Bengaluru, India

August 2023 - April 2025

- **LLM Task Processing Pipeline:** Built Redis and RQ-based worker pipelines for scalable LLM task execution, prompt processing, and asynchronous job orchestration across large API payloads.
- **Multi-Format API Parser and Test Generator:** Built a unified service-definition DSL for Postman, OpenAPI/Swagger, GraphQL, and Protocol Buffers, then integrated LLM APIs to generate test cases automatically, reducing test authoring time.
- **Visual API Testing Platform:** Built an API testing and execution platform with 50+ workflow node types for end-to-end orchestration, improving large-flow execution performance and reducing test runtime.
- **Browser Test Automation Tool:** Developed Chrome extension that records user interactions and generates Playwright test scripts. Implemented robust selectors with fallback strategies for UI changes. Achieved 95% stability, 3x test coverage, and 70% less manual QA.
- **Distributed API Workflow Execution Engine:** Built a distributed test execution engine that modeled large API collections as DAGs, executing nodes via Redis-backed worker queues (RQ) with dependency-aware scheduling, parallel fan-out, and isolated execution pipelines to support high-concurrency workflow simulation across 1000+ businesses.
- **Test Specification Compiler:** Built a compiler that normalized Postman, OpenAPI, GraphQL, and Protobuf specifications into a common executable graph IR, enabling a shared runtime for test generation, validation, and large-scale API workflow execution.

Castler

Full Stack Developer

Delhi, India

May 2021 - August 2023

- **Payment Processing Platform:** Built an event-driven payment platform for UPI/IMPS using message queues, idempotency controls, and circuit breakers. Processed INR 50Cr+ monthly across 2M+ transactions with sub-200ms latency while reducing infrastructure cost 35%.
- **Fault-Tolerant Payee Validation System:** Architected a fault-tolerant payee validation pipeline that absorbed 1000+ RPM peak demand against a heavily rate-limited upstream bank API, using asynchronous queueing, deduplication, backpressure, and stateful retries to avoid user-facing failures and preserve a seamless validation experience during rush-hour traffic.
- **Large-Scale Database Migration:** Led a zero-downtime migration of 200GB of financial data from MongoDB to MySQL using blue-green deployment and dual-write architecture. Built Airflow reconciliation pipelines that improved data consistency, delivered 40% faster queries, and reduced infrastructure cost 25%.
- **Security Automation and Compliance:** Implemented OAuth 2.0/PKCE authentication and automated security scanning with SonarQube, Snyk, and OWASP ZAP to strengthen application security and release hygiene.
- **Financial Consistency and Reconciliation:** Built automated state-reconciliation and recovery workflows across gateway responses, asynchronous bank callbacks, and internal ledger state to detect mismatched transaction states, recover failed payment flows, and preserve financial correctness at scale.

SKILLS

Languages: Python, TypeScript, Go, SQL

Backend & APIs: Node.js, NestJS, REST APIs, GraphQL, Microservices, Event-Driven Architecture, Distributed Systems

Data & Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Kafka, Airflow, AWS, Docker, Kubernetes, CI/CD

AI/ML: LLM APIs, RAG, Vector Databases

EDUCATION

Amity University

Bachelor of Technology in Computer Science

Lucknow, India

Aug. 2017 - May 2021